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A Contrastive Learning-Enhanced Residual Network for Predicting Epileptic Seizures Using EEG Signals

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The models used to predict epileptic seizures based on electroencephalogram (EEG) signals often encounter substantial challenges due to the requirement for large, labeled datasets and the inherent complexity of EEG data, which hinders their robustness and generalization capability. This study proposes CLResNet, a framework for predicting epileptic seizures, which combines contrastive self-supervised learning with a modified deep residual neural network to address the above challenges. In contrast to traditional models, CLResNet uses unlabeled EEG data for pre-training to extract robust feature representations. It is then fine-tuned on a smaller labeled dataset to significantly reduce its reliance on labeled data while improving its efficiency and predictive accuracy. The contrastive learning (CL) framework enhances the ability of the model to distinguish between preictal and interictal states, thus improving its robustness and generalizability. The architecture of CLResNet contains residual connections that enable it to learn deep features of the data and ensure an efficient gradient flow. The results of the evaluation of the model on the CHB-MIT dataset showed that it outperformed prevalent methods in the field, with an accuracy of 92.97%, sensitivity of 94.18%, and false-positive rate of 0.043/h. On the Siena dataset, the model also achieved competitive performance, with an accuracy of 92.79%, a sensitivity of 91.47%, and a false-positive rate of 0.041/h. These results confirm the effectiveness of CLResNet in addressing variations in EEG data, and show that contrastive self-supervised learning is a robust and accurate approach for predicting seizures.

Keywords: Epileptic seizure prediction; EEG; contrastive learning; improved ResNet.

1. Introduction

Epilepsy is a common neurological disorder that is characterized by abnormal neuronal firings that lead to recurrent seizures that are often unpredictable. These episodes not only compromise the patients' quality of life, but also pose serious, and sometimes life-threatening, risks.[1] While medical and surgical interventions can control seizures in many patients, approximately one-third of them remain resistant to currently available treatments, this highlights the critical need for alternative approaches.[2] The early prediction of seizures provides the opportunity for timely interventions through preventive measures to mitigate their physical, psychological, and social consequences.

Recent advances in Computer Science and Biomedical Engineering have accelerated research on

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the prediction of seizures based on electroencephalogram (EEG) technology. EEG, which records real-time brain activity, provides a valuable platform for identifying the precursory patterns that precede seizures.[3] However, the inherent complexity and inter-individual variability of EEG signals pose significant challenges to such identification. Traditional methods of signal processing often struggle to manage these complexities, which underscores the need for methods that can ensure high predictive accuracy and robustness. Machine learning (ML) and deep learning (DL) have lately emerged as transformative tools in this domain,[4] as they enable automated feature extraction from large-scale EEG data to facilitate the development of sophisticated predictive models.[5] Gu and Adeli proposed an EEG-based automated model for assessing the risk of sudden unexpected death in epilepsy (SUDEP).[6] Perez-Sanchez *et al.* developed a seizure prediction model leveraging ECG signals, expanding the scope beyond conventional EEG-based approaches.[7] Such approaches excel in terms of handling high-dimensional data, identifying intricate patterns, and capturing EEG Signal complexity and dynamics. Despite their potential, however, the issues of data imbalance, variations in EEG signals across patients, and the computational demands of real-time processing pose significant challenges to such approaches, and addressing them requires further research and methodological refinement.[8]

Advancements in computational algorithms have been largely driven by deep neural networks.[9,10] DL methods can reduce the need for manual interventions and accurately extract features from complex EEG signals to predict epileptic seizures. Many studies on seizure prediction have used CNN methods.[11,12] Truong *et al.* proposed a three-block CNN structure containing normalization, convolutional, and max-pooling layers,[13] while Ozcan *et al.* spatially arranged electrode channels from the CHB-MIT dataset to construct 3D inputs that were processed through four convolutional and three fully-connected layers.[14] However, the traditional CNN frequently fail to model extended data correlations,[15] and is susceptible to the problem of the vanishing gradient that afflicts deep networks. In recent years, Zhang *et al.* proposed that the combination of Transformer and convolutional networks can effectively extract information related to epilepsy seizures.[16,17] Shanir *et al.* developed a seizure detection method using 1D-LBP and KNN,[18] whereas Nogay and Adeli employed transfer learning with a pre-trained AlexNet on EEG-derived spectrograms.[19] ResNet is a robust alternative to the CNN that can ensure a stable gradient flow and improved predictive performance by the model.[20]

Originally designed for image classification, ResNet has been successfully adapted to such domains as medical signal processing.[21] Its use of residual connections enables efficient feature extraction across layers, thus making it ideal for application to high-dimensional time-series data, like EEG data. Moreover, the performance of ResNet improves with the size of the dataset, which allows it to fully exploit the growing availability of digitized EEG data. Singh and Lobiyal suggested a hybrid ResNet50-LSTM model for predicting seizures,[22] while Jiang *et al.* developed TASM_ResNet by integrating a temporal attention module with a pre-trained ResNet.[23] Both models achieved high accuracy on intracranial EEG signals to highlight the potential of ResNet in this field.

Contrastive learning (CL) has gained significant attention in recent research. Along with other self-supervised learning methods, it has become a prevalent approach across fields ranging from computer vision and natural language processing to medical signal analysis.[24] CL outperforms supervised learning in extracting the features of EEG data against the backdrop of representation learning. Feng *et al.* proposed a semantically attentive CL strategy (C-SCL) to address discrepancies between EEG signals and their textual representations.[25] Chang *et al.* developed DSSNet to generate meaningful multi-view representations to reduce dependence on labeled data while enhancing the accuracy of automatic sleep staging.[26] Zhao *et al.* enhanced the classification accuracy of epileptic seizures by employing a model that incorporated supervised contrastive loss at the projection layer alongside cross-entropy loss at the classification layer.[27] Similarly, Guo *et al.* leveraged CL to integrate cross-subject EEG features and improve the accuracy of prediction of seizures.[28] However, these methods rely on supervised CL based on labeled data, which makes seizure prediction problematic in scenarios in which labeled data are scarce or altogether unavailable.

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This study proposes a model for epileptic seizure prediction, named CLResNet, which integrates CL with residual networks. CL is used to construct similar and dissimilar pairs of samples to accurately learn the features of the data without requiring an extensive labeled dataset. It can capture subtle differences between normal and abnormal patterns of EEG activity even with a limited number of labeled samples, where this enhances both the discriminative capability and generalizability of the model. Owing to their deep architecture and robust capability of feature extraction, residual networks enable the precise identification of critical features in complex, temporal EEG signals. By combining the strength of self-supervised CL with the capabilities of deep representation of residual networks, CLResNet surpasses baseline models in terms of the accuracy of its predictions of epileptic seizures.

The structure of this paper is as follows: Section 2 introduces the database and methodology utilized in this study. Section 3 presents the experimental results, comparing the proposed method with existing approaches in the field. In Sec. 4, the performance of the proposed model is analyzed and discussed in relation to previous studies. Lastly, Sec. 5 summarizes the key findings and suggests potential directions for future research.

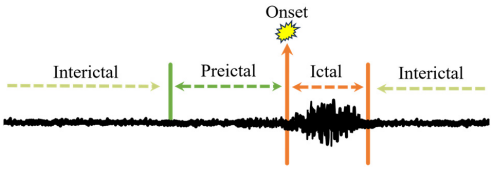

Fig. 1. Definitions of preictal, ictal, and interictal periods.

2. EEG Datasets

The CHB-MIT Scalp EEG Database and the Siena scalp EEG database were used in this study. The CHB-MIT Scalp EEG Database is a publicly available dataset, which is a widely recognized resource for seizure detection and prediction research.[3] Therefore, this study primarily conducts experiments using this dataset. The dataset comprises EEG recordings from 23 epilepsy patients. The recordings were collected following the international 10–20 system for electrode placement, with signals digitized at a sampling rate of 256 Hz. In addition, the Siena Scalp EEG dataset is also applied to further demonstrate the generalizability of the proposed method, which includes EEG recordings from 14 epilepsy patients. The EEG signals were recorded using the international 10–20 electrode system with a sampling rate of 512 Hz. In total, we conducted seizure prediction experiments on 181 seizure events and over 500 ho of EEG recordings from both datasets.

As shown in Fig. 1, the preictal period is defined as the 30 min preceding seizure onset, while the interictal period refers to the remaining time period excluding the ictal and preictal interval. For some patients, some seizures occurred in close succession with the preictal data being less than 30 min, and thus adjacent seizure events were merged into one seizure event for analysis. The specifics of the two EEG datasets utilized in this research are outlined in Table 1.

Table 1. Details of the CHB-MIT dataset and Siena dataset used in this study.

Dataset	Patient	Duration of interictal (h)	Duration of preictal (h)	Number of used seizures
CHB-MIT	chb01	22.80	3.50	7
	chb02	6.71	1.50	3
	chb03	6.00	3.00	7
	chb04	22.25	2.00	4
	chb05	18.06	2.50	5
	chb06	40.35	3.96	9
	chb07	14.25	1.50	3
	chb08	19.66	2.50	5
	chb09	16.78	1.50	4
	chb10	20.57	3.37	7
	chb11	1.80	0.89	3
	chb12	14.03	3.25	7
	chb13	14.73	3.00	8

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Table 1. (*Continued*)

Dataset	Patient	Duration of interictal (h)	Duration of preictal (h)	Number of used seizures
	chb14	16.15	3.74	8
	chb15	22.67	3.50	7
	chb16	9.42	1.59	6
	chb17	6.85	1.50	5
	chb18	5.47	2.48	6
	chb19	1.92	1.08	3
	chb20	3.62	3.53	8
	chb21	7.48	2.00	4
	chb22	12.58	1.35	3
	chb23	9.08	3.28	5
	chb24	13.13	7.00	8
	Total	326.36	63.52	135
Siena	PN00	1.28	1.21	4
	PN01	10.35	1.00	2
	PN03	15.00	1.00	2
	PN05	4.48	1.50	3
	PN06	9.53	2.50	5
	PN07	5.08	0.50	1
	PN09	5.33	1.50	3
	PN10	16.85	5.00	10
	PN11	1.90	0.50	1
	PN12	3.42	1.58	4
	PN13	7.15	1.50	3
	PN14	14.80	2.00	4
	PN16	3.82	1.00	2
	PN17	4.10	1.00	2
	Total	103.09	21.79	46

3. Method

The proposed method comprises three key components: pre-processing, the proposed CLResNet model, and post-processing. We describe each component in detail in the following sections.

3.1. *Pre-processing*

Two bandstop filters were used to reduce interference in specific frequency ranges due to noise from the power supply and other artifacts. The first filter targeted frequencies in the range of 117–123 Hz, while the second targeted those in the range of 57–63 Hz, and this reduced noise from power sources operating at 60 Hz. These frequency ranges were selected to suppress the power line interference and its first harmonic, which commonly appear in EEG signals recorded in regions with 60 Hz electricity. The slightly wider bandstop design helps eliminate spectral leakage and improves overall signal quality. To further enhance the quality of the signals, a 1 Hz high-pass filter was implemented to remove low-frequency elements, such as baseline drift and DC offsets. This process focused on high-frequency variations, which are more relevant when detecting epileptic seizures because low-frequency variations are typically not correlated with seizure-related neural activity.

A major challenge in predicting epileptic seizures is data imbalance, as seizure durations in the dataset are often considerably shorter than the interictal periods, resulting in a substantial disparity in the number of data points across categories. This can adversely affect predictive performance. To address the issue, we employed continuous and overlapping sampling techniques. Let L represent the size of the window for overlapping samples, M represent preictal signal duration, while N represents interictal signal duration. The number of segments generated for the preictal period after oversampling can then be calculated as $(M-L)/L+1$, while that for the corresponding interictal period is $(N-L)/L+1$.

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We set $L = 4\,\mathrm{s}$ in this experiment. If the interictal data still outweighed the preictal data during the training of the model, a portion of the former was randomly discarded to ensure a balanced ratio between the categories.

3.2. *CLResNet*

The proposed CLResNet framework for seizure prediction, illustrated in Fig. 2, comprises three key stages: Pre-training, fine-tuning, and testing. During the pre-training stage, raw EEG signals are augmented to generate diverse representations that are then processed by an improved ResNet-based feature extractor to learn high-dimensional embeddings. The model is optimized based on contrastive loss by enabling it to distinguish between positive and negative pairs of samples to extract meaningful and robust features. During the fine-tuning stage, the model undergoes additional training using labeled EEG data to classify the interictal and preictal states by minimizing cross-entropy loss and ensuring task-specific feature refinement. In the testing stage, the fine-tuned and improved ResNet processes EEG signals to extract the features, which are subsequently analyzed by a post-processing module to generate probability distributions of seizures over time. If a seizure is predicted, an alarm is triggered to enable timely intervention. This framework achieves enhanced performance and reliability in the context of seizure prediction by integrating CL with fine-tuning.

3.2.1. *Data augmentation*

Data augmentation plays a central role in CL, and enables the model to identify intrinsic data structures without labels by creating diverse "views" of each sample. The classic method of CL, SimCLR, focuses on transformations such as color adjustments and flipping in image-related data to generate pairs of positive samples from different perspectives.[29] By contrast, Qin *et al.* introduced the spatial variation generation algorithm, which improves data diversity by enhancing the density of sample neighborhoods, thereby improving the robustness of models used to classify motor imagery.[30] The spatial variation generation approach is better suited to capture spatial structural variations in EEG signals. It enables accurate simulations of subtle changes in electrode placement and the process of signal acquisition. In light of the characteristics of EEG data, we propose the following data augmentation strategies. Figure 3 presents sample transformed signals.

Rotation: Random rotations involve applying random angular transformations to EEG signals in the time domain to simulate signal variations caused by changes in head posture or electrode orientation.

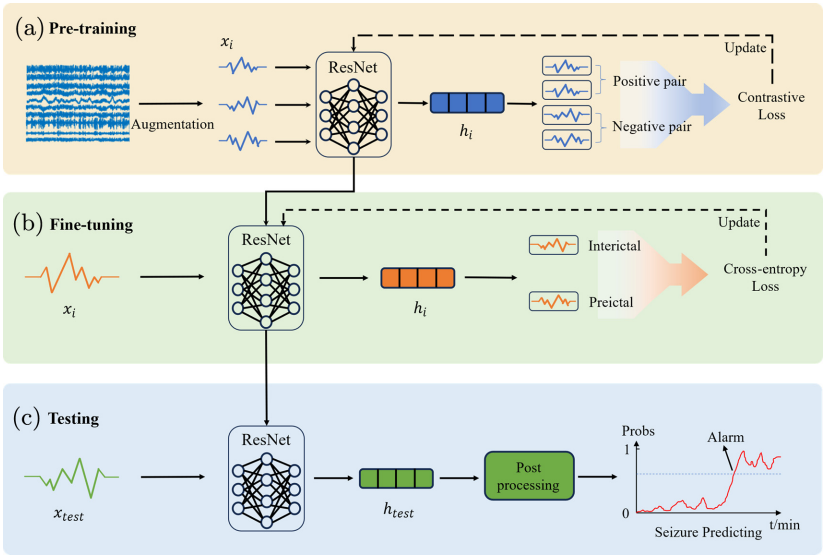

Fig. 2. The proposed CLResNet.

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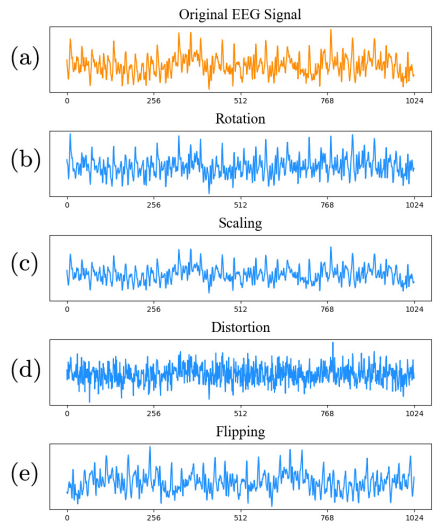

Fig. 3. Visual comparison of EEG signal augmentation method: (a) Original EEG signal, (b) rotation, (c) Scaling, (d) distortion, and (e) flipping.

The rotation angles are sampled from a normal distribution with a mean of 0 and a standard deviation of 0.1, and are respectively applied to sine and cosine transformation terms. This allows the augmented signals to exhibit variations in both amplitude and phase within a reasonable range. This strategy introduces diverse signal morphologies, thereby enhancing the model's robustness to spatial variations and electrode placement inconsistencies.

Scaling: Signal scaling aims to simulate amplitude fluctuations in EEG signals resulting from differences in sampling frequency or physiological variations across subjects. During augmentation, each signal is multiplied by a randomly sampled scaling factor within the range of [0.7,1.3], generating new samples that vary in amplitude while preserving their temporal structure. This enhances the model's ability to generalize across signals with differing amplitude distributions.

Distortion: The method is implemented by introducing Gaussian noise with a mean of 0 and a standard deviation ranging from 0.01 to 0.05 into the original signals. This simulates common types of interference encountered during EEG acquisition, such as electromyographic noise, power line interference, and electrode contact artifacts. This augmentation improves the model's robustness in low signal-to-noise ratio conditions and helps maintain predictive accuracy when processing noisy inputs.

Flipping: Flipping augmentation is performed by randomly mirroring EEG signals along the channel dimension to simulate spatial distortions caused by minor electrode placement shifts, which are commonly observed in real-world EEG recordings. This operation introduces spatial variability while preserving the underlying seizure-related patterns, thereby maintaining semantic consistency essential for CL. By increasing data diversity and enriching the construction of positive and negative pairs within the CL framework, this augmentation enhances the quality of learned representations and improves the model's generalization performance.

Pairs of positive samples are generated during training to represent diverse perspectives by randomly applying the above strategies for data augmentation. Pairs of negative samples are constructed by randomly selecting other samples from the batch.

3.2.2. *Contrastive loss function*

The contrastive loss is utilized to optimize the model's capability for feature extraction during training. This loss function aims to bring the representations of augmented views of the same signal closer together while pushing those of different signals farther apart. Before calculating the similarity between the signal representations, we standardize the input signals to guarantee that the magnitude of each feature vector is normalized to one. This eliminates the influence of the scale of the features on the calculation of similarity. We compute the cosine similarity of each pair of samples (positive or negative). The cosine similarity of two input representations $x_i^{(1)}$ and $x_j^{(2)}$ is expressed as follows:

$$\mathrm{sim}(x_i^{(1)}, x_j^{(2)}) = \frac{x_i^{(1)} \cdot x_j^{(2)}}{\|x_i^{(1)}\| \|x_j^{(2)}\|}. \tag{1}$$

Cosine similarity measures how similar two vectors are by calculating the angle between them. Greater similarity scores indicate proximity in feature space between compared signals. To prevent the model from comparing a sample with itself during training, we introduce a mask, with values of one on the diagonal and zero elsewhere, to ensure that each sample is compared only to other samples. The loss function enhances the model's performance by

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utilizing cross-entropy loss:

$$\text{Loss} = \frac{1}{N}\sum_{i=1}^{N} -\log \frac{\exp\left(\frac{\text{sim}(x_i^{(1)}, x_j^{(2)})}{\tau}\right)}{\sum_{j=1}^{2N} l_{[k\neq i]} \exp\left(\frac{\text{sim}(x_i^{(1)}, x_j^{(2)})}{\tau}\right)}, \quad (2)$$

where $x_i^{(1)}$ and $x_j^{(2)}$ represent two enhanced views of signal i, sim ( ) signifies the cosine similarity, τ denotes temperature, modulating distribution smoothness within CL frameworks. The numerator in the above formula represents the similarity between pairs of positive samples, while the denominator captures the similarity values across all pairs. The aim of optimizing the loss function is to enhance the similarity of positive sample pairs while increasing the separation between negative sample pairs. This allows the model to really get a handle on the underlying signal representations.

3.2.3. *ResNet architecture*

An advanced DL model based on an improved ResNet architecture is introduced, which in turn is based on the ResNet model developed by Hannun *et al.*[31] The model is designed with four main components: An input and feature extraction layer, a residual module, a CL module, and a classification module. By leveraging multi-scale convolutional kernels, residual connections, and CL strategies, can accurately capture the features of EEG signals to ensure a high accuracy of classification and robustness. The details are shown in Fig. 4(a).

The input and feature extraction layer processes multi-channel EEG signals to extract local temporal features from them by using a convolutional layer that is optimized for time-series data. This layer is succeeded by batch normalization and ReLU activation, which stabilize the training process and enhance the expressiveness of the extracted features. This design lays the foundation for the subsequent modules by preserving crucial patterns in the signals while reducing computational complexity.

At the heart of the model lies the residual module, which comprises four layers, each containing two residual blocks, as illustrated in Fig. 4(b). These blocks contain multi-scale convolutional kernels that enable the model to capture diverse spatiotemporal features at various resolutions. Table 2 presents the specifics concerning the model's variables. The parameters used in the proposed model was determined based on a combination of empirical tuning and insights from previous studies in EEG-based DL. The kernel sizes, strides, and paddings for each convolutional layer were selected according to standard practices commonly adopted in related research. The residual connections solve the problem of the vanishing gradient and enable the effective training of the network even as its depth increases.

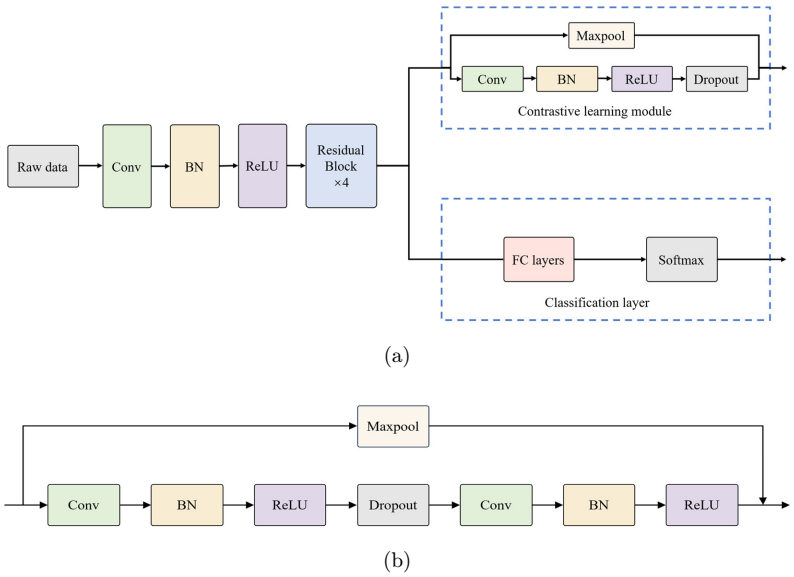

Fig. 4. (a) The structure of the proposed ResNet model. (b) Details of the proposed Residual Block.

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Table 2. Key parameters and the size of output of each model block.

Block name	Output size	Parameters
Input layer	$1 \times 18 \times 1024$	$k = (1, 9)$, $s = (1, 2)$, $p = (0, 4)$
Residual layer 1	$16 \times 64 \times 32$	$k_1 = (1, 15)$, $k_2 = (15, 1)$, $s_1 = (1, 2)$, $s_2 = (2, 1)$
Residual layer 2	$32 \times 32 \times 16$	$k_1 = (1, 11)$, $k_2 = (11, 1)$, $s_1 = (1, 2)$, $s_2 = (2, 1)$
Residual layer 3	$64 \times 16 \times 8$	$k_1 = (1, 7)$, $k_2 = (3, 1)$, $s_1 = (1, 2)$
Residual layer 4	$128 \times 8 \times 4$	$k_1 = (1, 3)$, $k_2 = (5, 5)$, $s_1 = (1, 2)$
Adaptive AvgPool	128	Global average pooling
Contrastive module	256	MLP: Linear ($128 \rightarrow 256$), Dropout ($p = 0.2$)
Classification	2	MLP: Linear ($128 \rightarrow 2$), Dropout ($p = 0.2$)

Notes: "Output Size" represents the dimensions of the map of the features output by each module. The parameters include the kernel size (k), stride (s), and padding (p).

As the features pass through the layers of the model, their number of dimensions decreases while the number of channels increases. This enables the extraction of abstract and high-level features.

After passing through the residual module, the extracted features are split into two paths. The first path leads to the CL module, which enhances the discriminative power of the feature representations. This module uses a series of fully connected layers and CL to distinguish between similar but distinct samples. The model achieves robust feature embeddings by maximizing positive pair relatedness while reducing negative pair relatedness. The second path connects to the classification module, which uses fully connected layers to refine the features for binary classification. Dropout regularization is applied to prevent overfitting, and the final layer outputs the predicted probabilities of the occurrence and nonoccurrence of a seizure by using a softmax function.

3.3. *Post-processing*

A moving average-based method was applied to smooth the one-dimensional time-series data, reducing noise and short-term fluctuations. This method involves sliding a window of a specified size over the data and computing the average value to generate a smoothened output. At its core, the process uses a convolution operation for averaging the data within the sliding window. To address edge effects, whereby the window cannot fully cover the data along the boundaries, a cumulative sum method is used along the boundaries to reduce distortion. This effectively reduces noise, stabilizes the data, and enhances their interpretability and reliability for subsequent analysis.

4. Experiments and Results

4.1. *Experimental settings*

To evaluate the performance and generalization capability of the proposed framework, experiments were carried out using Python 3.7, and PyTorch 1.10.0. A leave-one-out cross-validation (LOOCV) approach was employed for this assessment. It was used to validate the model against all possible combinations of the training set to ensure a robust assessment of its performance. The Adam optimizer was used for parameter optimization. All experiments were conducted on a workstation equipped with a 12th Gen Intel Core i7-12700K CPU (3.61 GHz), 64 GB RAM, and an NVIDIA GeForce RTX 3080 Ti GPU.

4.2. *Evaluation indicators*

In this work, two kinds of evaluation metrics were utilized to thoroughly evaluate the performance of the model: The segment-based metrics and the event-based metrics.

The first type of evaluation metrics based on the predicted segments consisted of a stringent set of metrics with accuracy (ACC), sensitivity (Sen), area under the curve (AUC), and p-value. ACC indicates the proportion of accurate predictions generated by the model, while Sen evaluates its ability to correctly identify the positive class of samples. The AUC reflects its overall performance across threshold

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settings, while the p-value evaluates the statistical significance of the model's outcomes. These metrics were calculated by using the following formulae:

$$\mathrm{ACC} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{FP} + \mathrm{FN} + \mathrm{TN}} \times 100\%, \tag{3}$$

$$\mathrm{Sen} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \times 100\%, \tag{4}$$

$$P\text{-values} = \sum_{i \geq m} \binom{i}{m} P^i (1-P)^{m-i}. \tag{5}$$

Here, each segment is independently classified as either preictal or interictal. True positive (TP) occurs when a preictal segment is correctly identified as preictal by the model, while false negative (FN) refers to a preictal segment incorrectly classified as interictal by model. Conversely, true negative (TN) denotes a correctly identified interictal segment, and false positive (FP) indicates an interictal segment mistakenly predicted as preictal.

The event-based evaluation metric is a measure to better represent the performance of the prediction algorithm for clinical application. The evaluation followed the definitions of the seizure occurrence period (SOP) and seizure prediction horizon (SPH) proposed by Maiwald *et al.*,[32] as shown in Fig. 5. After a prediction alarm, if a seizure event does not occur during the SPH, but during the SOP, the alarm is considered to be a valid prediction alarm. In clinical practice, the SPH represents the interval during which clinicians can intervene and implement preventive measures to mitigate seizures. Also, excessively long and SOP and SPH may cause unnecessary anxiety and tension for the patient. Based on this consideration, the SPH was set to 1 min, and the SOP was set to 30 min in our experiments.

In this work, when eight or more consecutive segments are classified as preictal by the model, we define this consecutive time interval as a definite

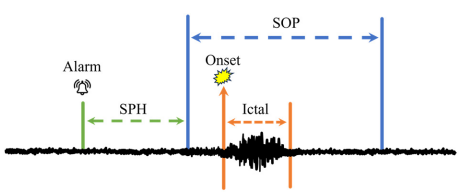

Fig. 5. A correct seizure event alarm based on the definitions of the SPH and SOP.

prediction event.[13] Thus, a correctly predicted seizure event means that the prediction event is identified and the seizure indeed occurs after the SPH and within the SOP. Otherwise, it is regarded as a false prediction event. Based on these statements, the event-based evaluation metric contains three indicators: Event-based sensitivity (S_n), false prediction rate (FPR), and average prediction time. S_n is calculated as the proportion of correctly predicted seizure events, that is, the number of correctly predicted seizure events divide by the total number of seizures as shown in formulae (6). FPR is measured as the number of false prediction event per hour. The average prediction time refers to the average time interval of the alarm prior to the beginning of seizure onset, which reflects the model's ability to provide early warning.

$$S_n = \frac{\text{The number of correctly predicted seizures}}{\text{The total number of seizures}} \times 100\%, \tag{6}$$

$$\mathrm{FPR} = \frac{\text{The number of false prediction events}}{\text{The hours of interictal}}. \tag{7}$$

4.3. *Overall performance*

The proposed model was evaluated on two widely used datasets: CHB-MIT and Siena. The experimental results are summarized in Table 3, which includes both segment-based and event-based performance metrics. The model achieved an average ACC of 92.97% and a Sen of 89.01% on the CHB-MIT dataset. These results demonstrate robust performance in predicting seizure events at the segment level, with an AUC of 0.927. In terms of event-based performance, the model achieved an average Sn of 94.18%, along with a low FPR of 0.043, indicating high precision and effectiveness in detecting true seizure events while minimizing false alarms. The prediction time averaged 19.62 min, showcasing the model's efficiency in real-time applications.

On the Siena dataset, the model achieved an average ACC of 92.79%, with a Sen of 91.47%, and an AUC of 0.922 at the segment level. Event-based results showed an average S_n of 91.19%, with a lower FPR of 0.041 compared to the CHB-MIT dataset, and a prediction time of 18.41 min. The Siena

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Table 3. Experimental results on the CHB-MIT dataset and Siena dataset.

		Segment-based level				Event-based level		
Dataset	Patient	ACC (%)	Sen (%)	AUC	*P*-value	S_n (%)	FPR/h	Prediction time (min)
CHB-MIT	chb01	99.59	98.28	0.999	< 0.001	100	0.000	22.89
	chb02	90.44	75.91	0.867	< 0.001	66.67	0.012	18.58
	chb03	93.17	86.1	0.964	0.004	83.33	0.265	19.39
	chb04	89.82	79.44	0.881	0.011	100	0.000	28.63
	chb05	93.53	83.29	0.948	< 0.001	100	0.000	13.77
	chb06	88.92	82.31	0.879	0.029	88.89	0.188	16.24
	chb07	92.00	81.40	0.914	0.032	100	0.000	23.97
	chb08	99.67	99.11	0.998	< 0.001	100	0.000	29.23
	chb09	92.99	87.33	0.916	0.005	75	0.121	15.84
	chb10	94.13	90.62	0.966	< 0.001	100	0.000	26.56
	chb11	93.39	94.35	0.947	< 0.001	100	0.000	9.13
	chb12	91.37	91.07	0.906	< 0.001	85.71	0.000	14.93
	chb13	94.33	96.08	0.935	< 0.001	100	0.003	18.58
	chb14	91.81	89.70	0.895	< 0.001	100	0.008	15.67
	chb15	89.24	88.50	0.876	< 0.001	85.71	0.021	12.50
	chb16	84.54	79.88	0.851	0.011	100	0.000	19.42
	chb17	99.76	99.01	0.999	0.011	100	0.000	28.68
	chb18	91.51	94.51	0.923	< 0.001	100	0.138	11.93
	chb19	91.02	90.78	0.918	0.036	100	0.000	19.59
	chb20	95.45	88.77	0.952	< 0.001	87.5	0.082	24.43
	chb21	91.34	84.82	0.899	0.021	100	0.000	14.35
	chb22	99.79	93.07	0.995	0.027	100	0.000	26.14
	chb23	98.45	98.69	0.975	< 0.001	100	0.000	22.15
	chb24	85.16	83.28	0.843	< 0.001	87.5	0.201	18.36
	Average	92.97 ± 4.24	89.01 ± 6.79	0.927 ± 0.048	—	94.18 ± 9.43	0.043 ± 0.078	19.62 ± 5.73
Siena	PN00	85.67	83.71	0.846	0.013	75	0.076	12.63
	PN01	89.15	84.52	0.867	0.024	100	0.059	15.28
	PN03	95.67	94.38	0.952	< 0.001	100	0.033	25.21
	PN05	86.95	86.29	0.873	< 0.001	66.67	0.043	18.10
	PN06	91.44	91.78	0.923	0.009	80	0.081	13.65
	PN07	98.34	98.86	0.982	< 0.001	100	0.024	28.22
	PN09	86.72	90.66	0.884	0.032	100	0.069	17.83
	PN10	96.35	94.52	0.945	0.020	80	0.036	16.58
	PN11	97.53	96.57	0.973	< 0.001	100	0.028	23.41
	PN12	93.49	94.35	0.926	< 0.001	100	0.012	13.16
	PN13	94.36	93.93	0.927	0.006	100	0.026	12.36
	PN14	94.86	86.92	0.958	< 0.001	75	0.045	19.34
	PN16	92.60	91.08	0.904	0.009	100	0.000	25.35
	PN17	95.94	93.07	0.949	0.003	100	0.045	16.61
	Average	92.79 ± 4.03	91.47 ± 4.41	0.922 ± 0.040	—	91.19 ± 12.17	0.041 ± 0.023	18.41 ± 5.02

dataset results indicate that the model performs consistently across different datasets.

4.4. *Impact of contrastive learning strategy*

To evaluate the contribution of CL to the performance of this seizure prediction model, an ablation study was conducted comparing the model with and without CL. The results are presented in Fig. 6 and Table 4, which show the performance differences in terms of ACC, AUC, Sen, FPR, and prediction time at both the segment-based and event-based levels.

At the segment-based level, the model with CL outperforms the model without CL in terms of all major metrics. Specifically, the ACC increased from 90.80% to 92.97%, and the AUC improved from

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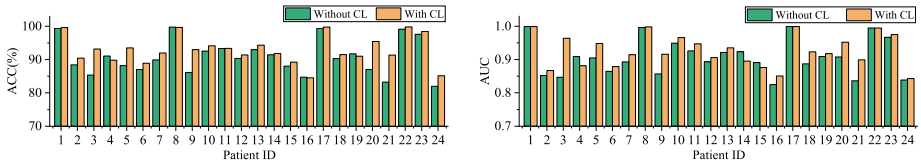


(a) (b)

Fig. 6. Comparison between models with and without CL on the CHB-MIT dataset. (a) ACC and (b) AUC.

Table 4. Comparative results of using CL on the CHB-MIT dataset.

	Segment-based level			Event-based level		
Method	ACC (%)	Sen (%)	AUC	S_n (%)	FPR/h	Prediction time (min)
Without CL	90.80 ± 5.23	87.30 ± 7.29	0.908 ± 0.054	89.43 ± 13.49	0.056 ± 0.095	18.46 ± 5.88
With CL	92.97 ± 4.24	89.01 ± 6.79	0.927 ± 0.048	94.18 ± 9.43	0.043 ± 0.078	19.62 ± 5.73

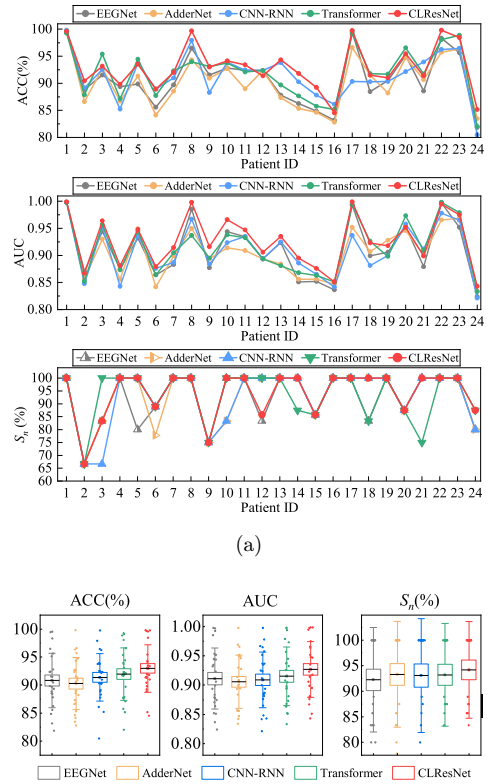
0.908 to 0.927. The Sen also showed a noticeable improvement, rising from 87.30% to 89.01%.

At the event-based level, the model with CL achieved a higher S_n of 94.18% compared to the model without CL, which had a Sn of 89.43%, indicating more precise detection of seizure events. Additionally, the FPR was lower for the model with CL at 0.043, compared to the model without CL at 0.056, reflecting an improved ability to minimize false alarms. Furthermore, its prediction time, which represents the temporal window preceding the prediction of a seizure, increased from 18.46 min to 19.62 min. This provides the patient and their medical team with more time to respond, and to implement interventions to mitigate or prevent the effects of the seizure.

4.5. *Apply contrastive learning to other baseline models*

To comprehensively evaluate the performance of the proposed model in seizure prediction tasks, four representative neural network architectures were selected for comparison: EEGNet, AdderNet, CNN-RNN, and Transformer. These models reflect recent advancements in temporal modeling, lightweight architectures, and attention mechanisms. Detailed results for each patient are presented in Fig. 7.

In terms of overall average performance, the CLResNet model achieved the best results across all



(a)

(b)

Fig. 7. Assessment of the proposed CLResNet in comparison with EEGNet, AdderNet, CNN-RNN, and Transformer on CHB-MIT dataset. (a) The three panels illustrate values of the ACC, AUC, and Sen of the three models for each patient (b). The three panels show their average values of ACC, AUC, and S_n.

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three metrics, with an average ACC of 92.97%, an AUC of 0.927, and a Sen of 94.18%. This demonstrates that the deep residual structure of CLResNet is effective in capturing complex EEG features. The Transformer model followed closely, with an average ACC of 91.97% and S_n of 93.21%, indicating its strong ability to model long-range temporal dependencies through global attention mechanisms.

EEGNet and AdderNet showed comparable performance, with average accuracies of 90.83% and 90.27%, respectively, and S_n of 92.24% and 93.31%. EEGNet, as a classic lightweight model for EEG processing, balances computational efficiency and accuracy. AdderNet replaces traditional multiplication operations with addition, significantly reducing computational cost while maintaining high accuracy. The CNN-RNN model achieved a slightly lower average ACC of 91.39% but maintained a S_n of 93.07%, suggesting strong recall capability in seizure prediction. This indicates that combining convolutional feature extraction with recurrent temporal modeling remains effective in EEG analysis.

4.6. *Effects of temperature*

The effect of the temperature-related parameter τ on the effectiveness of CL was examined. Experimental data were collected from three randomly selected patients, and the average results were used for analysis. Four values of τ were tested: 0.05, 0.1, 0.2, and 0.3. The selection of these specific τ values was guided by prior works in CL such as SimCLR[29] and MoCo,[33] where τ typically ranges between 0.05 and 0.3. Based on these references, representative values within this interval were chosen to evaluate the sensitivity of the model under different temperature settings. The evaluation focused on three key metrics: ACC, AUC, and prediction time.

In seizure prediction experiments based on CL, $\tau = 0.1$ was found to yield the best overall performance. As shown in Fig. 8(a), the highest ACC of 91.14% was achieved at $\tau = 0.1$, while lower ACC were observed at $\tau = 0.05$, 0.2, and 0.3, corresponding to 90.94%, 90.62%, and 90.01%, respectively. Similarly, Fig. 8(b) shows the highest AUC of 0.904 at $\tau = 0.1$, with reduced values at other settings. Figure 8(c) indicates that the prediction time was longest at $\tau = 0.1$, reaching 22.2 min. Although a slight increase was observed at $\tau = 0.3$, the prediction time remained shorter than at $\tau = 0.1$. These findings suggest that τ exerts a nonlinear effect on model performance: Smaller values impair feature discrimination, while larger values reduce sensitivity to fine-grained patterns.

5. Discussions

5.1. *T-SNE visualization*

T-SNE was used for visualization, taking patient chb02's data as an example, as shown in Fig. 9.

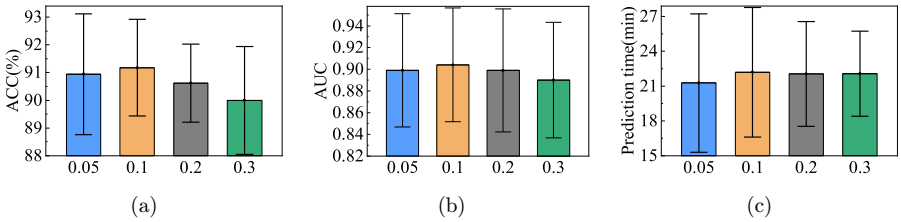

Fig. 8. Comparison of ACC, AUC, and prediction time with different temperature parameters (τ). (a) Average ACC (%) at different τ. (b) Average AUC at different τ. (c) Average prediction time at different τ.

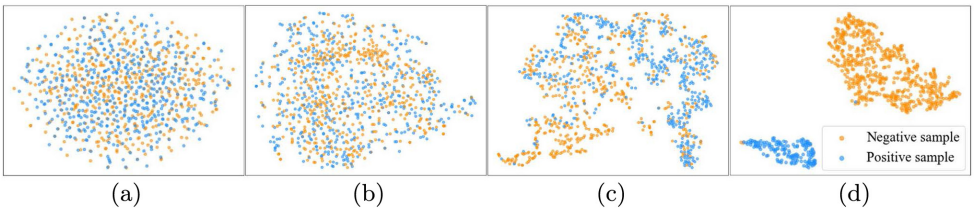

Fig. 9. *T*-SNE visualization of EEG data from patient 02: (a) Original data, (b) augmented data, (c) data used for pre-training, and (d) data used for fine-tuning.

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The significant overlap between the interictal (orange) and preictal (blue) states in the raw EEG signals indicates that distinguishing between these signals was challenging without feature optimization. A partial separation between the interictal and preictal data emerged after data augmentation, which shows that augmentation helped the model extract more discriminating features than before. Following pre-training based on CL, the distinction between these two states became clearer owing to a reduced overlap and better clustering. This demonstrates the improved ability of the model to identify pre-seizure states. Finally, training it on the specific task led to a nearly complete separation, highlighting its strong classification performance as well as its potential for reducing FPs and missed seizures. This progressive enhancement, achieved through data augmentation and CL, significantly strengthened the capability of the model to predict seizures, such that it can provide valuable early warning for patients and healthcare providers.

5.2. *Efficacy comparison of different models*

Table 5 compares the compactness and efficiency of several models in terms of parameters, FLOPs, convolution energy, and inference time. Among them, EEGNet is the most lightweight, with a low parameter count and fast inference speed. AdderNet shows the lowest energy consumption but suffers from high latency. CNN-RNN and Transformer models have high complexity and long inference times, making them less suitable for real-time tasks.

Although CLResNet has a relatively large number of parameters, it achieves the fastest inference time and maintains acceptable energy consumption. Compared to other high-performance models, CLResNet offers a better balance between accuracy, efficiency, and latency. For this reason, CLResNet was selected as the final model, as it meets the practical demands of real-time seizure prediction while maintaining strong predictive performance.

5.3. *Comparison of performance with different methods*

Table 6 compares the proposed CLResNet model with several recently developed models of seizure prediction. The findings underscore the exceptional performance of CLResNet, which leverages a contrastive self-supervised approach to learning. Unlike the other models, CLResNet was pre-trained on unlabeled data, and this enabled it to learn robust feature representations. This was followed by its fine-tuning with a smaller amount of labeled data, which allowed the model to achieve high predictive performance with few labeled samples — a significant advantage over traditional approaches based on supervised learning. Compared with previous studies, the proposed CLResNet achieved better performance on both the CHB-MIT and Siena datasets.

The proposed model achieved an average AUC of 0.927, a sensitivity of 94.18%, and the lowest FPR among all compared models, at 0.043/h. Although FB-CapsNet[38] recorded a slightly higher AUC of 0.948 and sensitivity of 95.7%, its FPR was significantly higher at 0.087. A low FPR is crucial for reducing false alarms in practical applications. Similarly, AddNet-SCL[27] attained comparable performance, with an AUC of 0.929 and a sensitivity of 93%, but exhibited a higher FPR of 0.094. These results demonstrate the capability of CLResNet to effectively balance accuracy and reliability.

Furthermore, the use of contrastive self-supervised learning in CLResNet enhanced its generalization ability across diverse patient data, outperforming CNN-based approaches proposed in

Table 5. Comparison of model compactness.

Method	Parameters ($\times 10^4$)	Flops ($\times 10^6$)	Convolution energy (mJ)	Inference time (ms)
EEGNet	4.55	33.89	0.228	2.82
AdderNet	11.91	79.04	0.117	33.44
CNN-RNN	46.61	408.14	0.112	5.77
Transformer	54.81	543.44	0.194	68.38
CLResNet	49.80	153.67	0.298	2.78

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Table 6. Comparison between the proposed model and prevalent methods of seizure prediction on the CHB-MIT and Siena dataset.

Author	Method	Dataset	Window length (s)	Preictal interval (min)	No. of patients	ACC (%)	AUC	Sen (%)	FPR (/h)
Usman *et al.*[34]	SVM	CHB-MIT	30	—	24	—	—	92.7	—
Yang *et al.*[35]	RDANet	CHB-MIT	5	30	13	92.07	0.893	91.3	—
Zhao *et al.*[27]	AddNet-SCL	CHB-MIT	4	30	19	—	0.929	93.0	0.094
Jemal *et al.*[36]	CNN inspired by FBCSP	CHB-MIT	5	30	23	90.9	0.918	96.1	0.04
Liang *et al.*[37]	CNN	CHB-MIT	30	30	13	—	—	88.3	0.04
Li *et al.*[38]	FB-CapsNet	CHB-MIT	4	60	19	—	0.948	95.7	0.087
Guo *et al.*[28]	CLEP-STS-Net	CHB-MIT	5	15	19	—	0.918	96.7	0.072
Assali *et al.*[39]	CNN	CHB-MIT	2	30	17	90.1	0.928	—	—
Jiang *et al.*[40]	Random forest	Siena	30	30	12	85.71	—	—	—
Li *et al.*[41]	Bi-LSTM	Siena	1 and 8	—	6	—	0.906	83.54	—
This work	CLResNet	CHB-MIT	4	30	24	92.97	0.927	94.18	0.043
		Siena			8	91.58	0.914	90.38	0.046

previous studies. For example, the model by Liang *et al.* resulted in a notably higher FPR of 0.3, while the method by Assali *et al.* showed inconsistent performance across patients due to its reliance on supervised training.[37,39] While Jemal *et al.* achieved a slightly higher sensitivity and a marginally lower FPR,[36] their model yielded lower accuracy and AUC compared to ours. Additionally, our study was conducted using data from all 24 patient cases, offering a more extensive and comprehensive evaluation. Moreover, based on the analysis of the computational efficiency of our model, our model provides a favorable trade-off among predictive performance, computational efficiency and universality, which is quite competitive.

5.4. *Limitations and future directions*

The experimental results showcase that the proposed model can perform well on data from different patients, and excels in particular in terms of its sensitivity and FPR compared with prevalent methods. Employing CL boosts its robustness in managing physiological differences among patients, which is important for clinical applications. However, it's crucial to mention that its predictive performance may be influenced by patient-specific factors and patterns of seizures. This highlights the need for future research to further refine individualized strategies for modeling. Moreover, although our model achieved commendable results in reducing the FPR, it requires further improvements. Future work should prioritize minimizing FPs while enhancing their capabilities of generalization across diverse scenarios and environments.

6. Conclusion

This paper presents a seizure prediction method using CL and a deep residual network to enhance accuracy by extracting multi-dimensional EEG features. Experiments demonstrate that the proposed model outperforms existing methods in accuracy, sensitivity, and FPR. A key advantage of our approach lies in its use of contrastive self-supervised learning, which enables the model to leverage unlabeled EEG data for pre-training. This significantly reduces its reliance on labeled data while improving its robustness and generalizability across patients. The application of CL enables the model to identify crucial distinctions between the preictal and interictal states, thus ensuring efficient feature extraction and sound predictive performance. Furthermore, this self-supervised approach to learning enhances the adaptability of the model, making it more effective in handling EEG signal variability. Future research in the area should focus on further optimizing the strategy for individualized modeling and reducing the FP rate of our model so that it can achieve even higher precision in clinical applications. These

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advances will pave the way for more reliable and practical systems to predict seizures that can improve clinical outcomes.

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